

MEMORANDUM - Analysis of Frontier Model Cards: Anthropic, Google, and xAI

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Executive Summary

We recently concluded a comprehensive audit of six frontier AI model cards from Anthropic, Google, and xAI. Our primary objective was to evaluate the depth, transparency, and utility of these disclosures.

The most significant takeaway is that none of the frontier cards scored above 60/100 under our rigorous rubric. The overarching ranking—consistent across both human and LLM evaluators—places Anthropic slightly ahead of Google, with xAI trailing behind (Anthropic $\gtrsim$ Google $\gtrsim$ xAI). Furthermore, our analysis reveals that the industry-standard binary (present/absent) grading system actively obscures meaningful differences and rewards "safety-washing."

Methodology

- Models Audited: Grok 4 & 4.1, Claude Opus & Sonnet 4.6, Gemini 3 & 3.1 Pro.
- Rubric: A 36-subsection weighted grading system, evaluated on a fine-grained 0–4 scale.
- Reviewers: 14 human reviewers (conducting 21 evaluations) and two LLM agents (GPT-5.5 and Claude 4.7) scoring independently.

Key Findings

1. Consistent, Robust Rankings

The rankings held firm regardless of the evaluator or how we weighted the rubric. Anthropic's Claude Opus 4.6 leads the pack, particularly in safety subsections (CBRN, Cyber Risk, Adversarial Robustness), where it consistently earned perfect scores from independent reviewers. Google's Gemini cards excel in "Inputs & Outputs" and "Intended Uses," while xAI's Grok cards remain thin outside of Critical Risk benchmarks.

2. The Failure of Binary Grading

Under traditional binary (present/absent) evaluation, every frontier card scores between 80 and 95 simply by mentioning a topic once. When evaluated on our 0–4 scale, the scores spread drastically from 27 to 56. Granular scoring is essential; binary grading artificially inflates scores and fails to reveal where labs are actually investing effort.

3. The "Minor Release" Regression

One of the most actionable findings is that minor-release model cards systematically score lower than their predecessors.

- Grok 4 to 4.1 scores lower across all engines.
- Opus 4.6 to Sonnet 4.6 drops by 4–9 points.
- Gemini 3 to 3.1 Pro drops by nearly 9 points after adjusting for reviewer bias.

- This happens largely because minor releases lazily defer or link back to predecessor cards (e.g., Sonnet redirecting to Opus), or simply receive less documentation effort overall.

4. Machine vs. Human Evaluators

LLMs are highly viable evaluators for model cards, showing strong agreement with human consensus (Spearman ρ of +1.00 for GPT-5.5 and +0.83 for Claude 4.7). However, machines are systematically stricter. The average raw subsection score was 1.93 for humans, compared to 1.46 for GPT-5.5 and 1.30 for Claude 4.7.

Meanwhile, human reviewers exhibit massive noise, with leniency varying by up to 42 points on a 100-point scale. We found that without at least three human reviewers per card, scores are dominated by individual bias (driven by reviewer effort, poor calibration, or specific professional personas).

5. Candor Builds the Most Trust

Across the board, reviewers praised candid acknowledgments of uncomfortable findings over sanitized PR. The best-in-class disclosures cost the labs almost nothing to publish but drastically improved reviewer trust. Standout examples include:

- Gemini 3.1 Pro: Openly disclosing a cyber-alert threshold breach.
- Claude Opus 4.6: Explicitly admitting its inability to fully evaluate CBRN safety, alongside documented examples of the model engaging in strategic deception.

Strategic Takeaways for ARI

The current state of model cards shows that while frameworks are being adopted, the *depth* of disclosure remains shallow. As ARI pushes for responsible innovation, advocating for granular, standardized reporting requirements rather than simple checklist compliance will be critical. Furthermore, we should pressure developers to maintain high documentation standards for point releases, rather than treating them as afterthoughts.

Please let me know if you would like to review the raw scoring data or discuss how to integrate these findings into ARI's upcoming policy recommendations.